THE FUNDAMENTAL SYSTEMS TENSION OF PHYSICAL AI

High-Capacity Learned Generative Models ⟷ Hard Real-Time Deterministic Safety Guarantees

HIGH-CAPACITY LEARNED MODELS (Cortex)

Transformers & Diffusion Policies

- **Generalization:**
  Open-world semantic reasoning, novel visual grounding
- **Computational Substrate:**
  Heterogeneous NPU / GPU with large DRAM footprint
- **Execution Paradigm:**
  Batched tensor inference, stochastic sampling
- **Latency Profile:**
  Variable 15–100 ms with long P99.9 tail spikes
- **Failure Characteristics:**
  Hallucinations, out-of-distribution drift, crashes

PROPOSE
⟷
PERMIT

HARD REAL-TIME DETERMINISTIC REFLEX (Enforcer)

Zero-Allocation Active-Set QP Barrier Solvers

- **Safety Guarantee:**
  Strict forward invariance $h(x) \geq 0$, zero collisions
- **Computational Substrate:**
  Real-Time MCU in static SRAM (0 dynamic malloc)
- **Execution Paradigm:**
  Deterministic 1000 Hz loop, bounded iterations
- **Latency Profile:**
  Bounded ≤ 175 μs with microsecond jitter (< 5 μs)
- **Failure Characteristics:**
  Fail-safe hardware arrest, Category 0/1 stops